

Cleaved hemagglutinin in ESM3:

Can the model treat HA1 and HA2 as one molecule joined by a disulfide bridge?

my_extendedSampling / softmax / bridge_test

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Abstract

`zero_polymer` is an influenza hemagglutinin precursor (HA0). In the mature protein it is cut once, after Arg329 in mature numbering, into HA1 and HA2, which stay together through an interchain disulfide between Cys14 of HA1 and Cys137 of HA2. The question was whether ESM3 can be given that two-chain molecule and handle it. The mechanical answer is yes: the stock tokenizer accepts the chain-break character, and presenting the chains as separate but connected measurably helps HA1. The structural answer is no, with caveats: no assembled construct folds well enough to pass the geometry gate, and the interchain Cys14–Cys137 distance is 96 Å (cleaved) and 130 Å (no break), against a bonding criterion of 4.5 Å. Two known-disulfide controls then showed that the distance criterion itself is weaker than I first said: it has some sensitivity but no specificity. This report records the results, the corrections made along the way, the limitations, and a prioritised to-do list.

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1 Summary

- **Mechanics work.** `model.encode(ESMProtein(sequence=...))` accepts |; 551 positions go in and 551 come out; `to_protein_complex()` recovers two chains. No change to the model code was needed.
- **The chain-break helps.** HA1’s mean pLDDT goes from 51.4 (fused, no break) to 63.2 (with break), and the pLDDT at its Cys14 anchor from 22.5 to 73.6. HA1 alone scores 59.8.
- **HA2 alone folds well; the assembly does not.** HA2 alone: pLDDT 93.9, no clashes, passes the gate. Every HA1-containing construct fails it.
- **The interchain bridge is not formed** in either assembled construct, by a margin (more than 20× the criterion) that no threshold subtlety can change.
- **The 4.5 Å criterion is not a reliable bond detector.** BPTI: 3/3 known bonds inside it. Crambin: 1/3 inside it, and the closest pair in the molecule is a non-native one at 3.33 Å.
- **Not tested:** the trimer, any experimental reference structure, step-matched comparisons, partner identity for HA.

2 What was built

All of it is on branch `claude/esm3-disulfide-polymer-chains-rwvpoq`.

Commit	Content
46bccd8	<code>tests/references.py</code> : chains, bridge, numbering helpers
1f35fbf, fcb86aa, caeb436	<code>tests/test_cleaved_complex.py</code> , three revisions
4056d91, 346b3de	BPTI and crambin controls; known-disulfide measurement
086e2cd, 1b0d3ce, 6b34313, 8efcd2c	<code>bridge_test/DEVLOG_bridge_test.txt</code>
87dc571	<code>bridge_test/DEVLOG_calibration.txt</code>

The construct. Chain one is mature HA1 (329 aa, ends ...QTR); chain two is HA2 (221 aa, starts GIFGAI); the 16-residue signal peptide is dropped, so 550 residues in total. Both chains are slices of `ZERO_POLYMER`, never retyped, and `_check()` asserts the junction, the lengths, and that both anchors are cysteines.

A numbering trap worth keeping. The HA landmarks are in *mature* numbering, which runs 16 behind the index into `ZERO_POLYMER`. Read naively, `ZERO_POLYMER[328]` (“Arg329”) is T and `ZERO_POLYMER[13]` (“Cys14”) is V. Neither raises. `mature_to_index()` and `chain_index()` exist so the conversion is never done by hand.

3 Method

Route into the model. The sampler’s usual path (`ExtendedProtein` with `predict_attention` on soft `sequence_probs`) cannot carry a chain break: `expand()` maps through a 25-letter vocabulary with no |, and the encoder zero-pads over token 31. The stock `encode/predict_protein` route runs the real tokenizer and has none of these problems. Two shortcuts that look right are wrong: `chain_id=` is consumed only by geometric attention, which the soft path masks and zeroes, so it is a silent no-op; and `sequence_id=` is an attention separator, so distinct ids would make the chains ignore each other.

What the decoder returns. Backbone plus an inferred $C\beta$, and no side-chain sulfur at any residue. So SG–SG is unavailable and $C\beta$ – $C\beta$ distance is the only disulfide proxy. $C\beta$ is not predicted: `infer_cbeta()` derives it from N/CA/C and writes NaN at every glycine. The chain-break position is excluded from all $C\beta$ statistics.

The instrument.

- *Geometry gate:* pTM ≥ 0.40 , and no $C\beta$ – $C\beta$ pair closer than 3.0 Å at sequence separation ≥ 2 .
- *Bond criterion:* $C\beta$ – $C\beta \leq 4.5$ Å.
- *Controls:* protein G (fold control, no cysteines); BPTI and crambin (three known disulfides each).

4 Results

4.1 The controls

Reference	steps	pTM	pLDDT	% ≥ 70	min $C\beta$ (Å)	clashes
protein G (56 aa)	56	0.7596	93.24	100.0	3.68	0
BPTI (58 aa)	58	0.7545	92.89	98.3	3.35	0
crambin (46 aa)	46	0.5816	83.31	89.1	3.33	0

All three pass the gate, so the harness folds, and the choice of 3.0 Å as the clash line is borne out empirically (confident folds have their closest pair at 3.3–3.7 Å).

4.2 HA constructs

	cleaved	mature	HA1 alone	HA2 alone
positions / steps	551 / 256	550 / 256	329 / 329	221 / 221
pTM	0.2322	0.2614	0.4133	0.5674
pLDDT, all residues	63.89	59.20	59.83	93.93
% residues ≥ 70	45.5	31.8	18.5	97.7
pLDDT, HA1 (% ≥ 70)	63.22 (38.6)	51.42 (11.2)	59.83 (18.5)	—
pLDDT, HA2 (% ≥ 70)	64.90 (55.7)	70.78 (62.4)	—	93.93 (97.7)
min $C\beta$ – $C\beta$ (Å)	1.50	1.17	0.44	4.14
clashes	15	9	16	0
gate	fail	fail	fail	pass
Cys14–Cys137 $C\beta$ (Å)	95.67	129.74	—	—
pLDDT at Cys14 / Cys137	73.6 / 61.3	22.5 / 80.9	—	—

The first run, at one decoding step, gave pTM 0.1827 and a bridge distance of 38.55 Å with a 1.64 Å closest cysteine pair. It is not tabulated because it measured a structure the model did not believe in; see Section 5.

Reproducibility. Cleaved and mature were each run twice, with different script versions, on the same machine. Both gave identical pTM (0.2322 and 0.2614) and identical bridge distances (95.67 and 129.74 Å). Decoding at temperature zero looks deterministic here. This is two repeat pairs on one machine, not a proof.

4.3 Findings

The chain-break helps HA1, most at the bridge. HA1 pLDDT rises $51.42 \rightarrow 63.22$ and its confident fraction $11.2\% \rightarrow 38.6\%$ when the break is declared. At the Cys14 anchor the rise is $22.5 \rightarrow 73.6$. HA2 pays a little ($70.78 \rightarrow 64.90$).

The clash *topology* differs. Mature has fewer clashes (9 against 15) but they sit at $|i - j|$ of 80–159, so distant segments are superimposed; cleaved’s are all local ($|i - j| \leq 16$). Counting clashes without their separation would rank the two backwards.

The two chains help each other asymmetrically. HA1 gains from HA2 (alone 59.83, in the cleaved complex 63.22), and is *worse* with HA2 fused (51.42) than with no HA2 at all. HA2 loses heavily to company (93.93 alone, 64.90 in the complex).

HA1 is the hard half in every construct. Alone, with a full step budget, it still fails the gate with 16 clashes and a minimum $C\beta$ separation of $0.44 \text{ \AA}$.

The bridge is not formed. $95.67 \text{ \AA}$ and $129.74 \text{ \AA}$ are about $21\times$ and $29\times$ the criterion; ranks 98 and 95 of 136 cysteine pairs. This holds under any reasonable criterion.

4.4 Disulfide calibration

Protein	Bond	$C\beta$ – $C\beta$ ($\text{\AA}$)	rank	within $4.5 \text{ \AA}$
BPTI	Cys14–Cys38	3.60	1/15	yes
BPTI	Cys5–Cys55	3.66	2/15	yes
BPTI	Cys30–Cys51	4.45	3/15	yes
crambin	Cys4–Cys32	3.79	2/15	yes
crambin	Cys3–Cys40	7.75	8/15	no
crambin	Cys16–Cys26	10.42	12/15	no
crambin	Cys16–Cys40 (<i>non-native</i>)	3.33	1/15	yes
BPTI	Cys51–Cys55 (<i>non-native, closest</i>)	6.15	4/15	no

BPTI separates cleanly: the last true bond is at $4.45 \text{ \AA}$ and the first false pair at $6.15 \text{ \AA}$. Crambin fails twice over. It misses two real bonds, and it puts a non-native pair inside the criterion. Partner recovery (known pairs among the three closest) is 3/3 for BPTI and 1/3 for crambin. A distance threshold is therefore necessary for a bond and not sufficient.

5 Corrections made during the session

These are recorded because several first readings were wrong.

1. **One decoding step.** The first verdict (“bridge not formed, $38.55 \text{ \AA}$ ”) came from a script that used the default `num_steps=1`, which samples all 551 structure tokens at once. It scored pTM 0.18 with a $1.64 \text{ \AA}$ “contact” shorter than a C–C bond. The verdict was worthless. The gate now runs before any verdict.
2. **A control that did not control.** Protein G validated the fold but has no cysteines, so it could not validate the measurement. BPTI and crambin were added for that reason.
3. **Two overstated claims, retracted.** The calibration script printed “ $4.5 \text{ \AA}$ is sound; HA2’s disulfides are not formed” from BPTI alone. That tested sensitivity on one protein and

specificity not at all. Separately, HA1-alone pairs at 3.51 and 3.92 Å had been described as “real disulfide geometry”; by the crambin result they could be non-native.

4. **A pre-registered prediction failed.** I expected the trimer signature (both chains confident, assembly low). The result was asymmetric instead: HA2 adequate, HA1 poor.
5. **Small bugs.** The chain-break position was admitted into the $C\beta$ statistics; labels were wrong for the break-free reference; a results file was overwritten by a generic filename.

6 Limitations

6.1 Of the measurements

- **No ground truth.** There is no experimental structure in the loop, so no RMSD or TM-score. pLDDT and pTM are the model’s own confidence; crambin’s pLDDT of 83 coexists with wrong partners.
- **$C\beta$ – $C\beta$ is a proxy.** The decoder emits no sulfur, and $C\beta$ is inferred from the backbone. A bond criterion built on it inherits any backbone error, and glycines (which have no $C\beta$) are invisible to the clash count.
- **Steps are not matched across constructs.** HA1 and HA2 alone used 329 and 221 steps; the assemblies used 256. Isolated chains therefore decode about one token per step and the assemblies about two. The HA1-alone against HA1-in-complex comparison is confounded by this, in a direction I have not measured.
- **One run per condition.** The two repeat pairs agree exactly, but no condition has been run across seeds or checkpoints, and nothing here gives error bars.
- **pTM depends on length,** so comparing it between a 46-residue and a 550-residue chain is weak. Only same-construct comparisons are clean.
- **Calibration is two proteins.** BPTI and crambin are small and well-ordered. Neither says what the criterion does in a 550-residue fold.
- **Sequences were supplied, not independently verified.** The BPTI and crambin sequences and pairings came with sources. I checked them for internal consistency (length, cysteine positions, one-to-one pairing, crambin’s Pro22/Leu25 isoform) but not against UniProt directly.
- **Only esm3-open.** No other checkpoint was tried.

6.2 Of the scope

- **Monomer only.** HA is a trimer. A single HA1+HA2 protomer may have no self-contained fold, and a low score may then be correct. Not tested.
- **No verified native disulfide list for HA,** so partner recovery cannot be run on HA at all. Positive bridge claims about HA are not possible yet.
- **The energy machinery cannot take a chain break.** The sampler’s soft-probability path has no way to carry token 31, so none of the U_{am} work has been applied to the cleaved form.
- **Glycans, the fusion peptide, and everything outside the sequence** are absent by construction.

6.3 Of the session itself

- For most of the session I could not run code: the shell was blocked until auto mode was switched off in this final turn. Every script was therefore written unexecuted and validated by your runs. One consequence: `test_cleaved_complex.py` still carries an unused `present_atoms()` helper that I reintroduced by mistake.
- One commit message (87dc571) has a stray line that refers to non-existent session notes. The file content is fine; the message cannot be amended without rewriting history.

7 To do

7.1 Recommended next step

Do the **step-matched rerun** first, because it is cheap and it is the one confound that touches a headline result:

```
python tests/test_cleaved_complex.py --ref one --steps 256
python tests/test_cleaved_complex.py --ref two --steps 256
```

Then compare HA1 alone against the cleaved HA1 (63.22) at equal step count. If HA1 alone is still lower, “HA2’s presence helps HA1” stands; if it catches up, the effect was partly the step budget.

7.2 Prioritised list

1. **Step-matched HA1 and HA2** (above).
2. **Per-residue pLDDT at the known-bond residues** in the BPTI and crambin runs. If crambin’s Cys3, Cys16, Cys26 and Cys40 are low-confidence, the misses are uncertainty and a pLDDT filter may repair the criterion; if confident, the placement is wrong. The calibration block does not print this yet.
3. **Report specificity in the calibration verdict:** the count of in-range non-native pairs next to the count of in-range known bonds.
4. **Experimental references for BPTI and crambin** (named PDB entries, checked against the sequences) and a per-residue C α RMSD. That turns “pLDDT 83” into a measured error.
5. **A verified native disulfide list for HA**, from a named structure entry, asserted at import the way the controls are. Required before any partner-recovery claim about HA.
6. **The trimer:** `one|two|one|two|one|two`, 1653 positions. The biologically correct question, and expensive.
7. **Condition on a real HA structure** and ask whether ESM3 *maintains* the bridge rather than discovers it. A different and more tractable question.
8. **Only if the cleaved form is needed in the energy machinery:** inject a one-hot at token 31 in the padded region, set the structure token at the break to `STRUCTURE_CHAINBREAK_TOKEN` in `_default()`, and exclude the break site from `mutate()`.

7.3 Housekeeping

- At session end, append the two devlogs to the main one:

```
cat bridge_test/DEVLOG_bridge_test.txt \  
    bridge_test/DEVLOG_calibration.txt >> DEVLOG.txt
```

- Give results files run-identifying names; the generic default overwrote the protein G control once.
- Remove the dead `present_atoms()` helper.

8 Reproducing the runs

From `softmax/code`:

```
python tests/test_cleaved_complex.py --ref protein_g --steps 56  
python tests/test_cleaved_complex.py --ref bpti      --steps 58  
python tests/test_cleaved_complex.py --ref crambin   --steps 46  
python tests/test_cleaved_complex.py --ref cleaved   --steps 256  
python tests/test_cleaved_complex.py --ref mature    --steps 256  
python tests/test_cleaved_complex.py --ref one       --steps 329  
python tests/test_cleaved_complex.py --ref two       --steps 221
```

Output files are in `softmax/code/bridge_test/`.